Comparative analysis — our CSI pilot vs published WiFi sensing datasets

Answers the question *"did we collect our data correctly?"* by measuring the pilot capture and comparing it against three published datasets (ARIL, UT-HAR, Widar 3.0).

Deliverable: `comparative_analysis.pdf` (19 slides, 16:9).

Headline findings

	Finding
✅	**Sampling rate is sufficient** — all class-discriminative energy lies below **18 Hz**; our worst-case Nyquist is 25.7 Hz. The 15× rate deficit vs UT-HAR costs nothing measurable *on this task*.
✅	**Signal is strong** — `action` carries **4.2×** the motion energy of `noaction` (Fisher ratio 3.85).
✅	**Our evaluation is the strictest** of the four datasets — leave-one-subject-out, vs ARIL's every-5th-sample and UT-HAR's random split.
✅	**Front-end matches ARIL** spec-for-spec (USRP N210, 52 subcarriers, SISO, amplitude-only).
⚠️	**The task is near-trivial** — one scalar + one threshold scores **99.1%**. Our 98.4% deep model does *not* beat that baseline.
⚠️	**13% of windows silently dropped** — `segment_all_pilot.py` used the default `--rate 100` on 66 Hz data. Fixable with `--rate 50`, no recapture.
⚠️	**~4 effective dimensions, not 52** — subcarriers are highly correlated during motion, undercutting ImgFi's "×52 free data" claim.
❌	**Scale/diversity** — 322 samples vs 1,394 / 4,477 / 258,000; 2 subjects, 1 room, 1 day.

Files

File	What
`comparative_analysis.tex` / `.pdf`	The deck (build: `pdflatex comparative_analysis.tex`, run twice)
`analyze_pilot.py`	Measures every pilot statistic → `pilot_stats.json`
`make_figures.py`	Builds all 10 figures → `figures/*.pdf`
`pilot_stats.json`	Machine-readable results (rates, jitter, correlation, effective rank, separability, window yield)

Rebuild

```
cd ~/usrp-projects/analysis
python3 analyze_pilot.py     # -> pilot_stats.json
python3 make_figures.py      # -> figures/*.pdf
pdflatex comparative_analysis.tex && pdflatex comparative_analysis.tex
```

Requires `numpy pandas scipy matplotlib` and a LaTeX install with `beamer`.

Scope limitation (stated on the last slide too)

Reference datasets are compared on **published specifications only** — we did not download and re-analyse their raw CSI, so their signal-level statistics (jitter, effective rank, spectral ratio) are not measured here. Every number describing *our* pilot is measured directly from `data/pilot/`.